

Background subtraction $M = S + B$: predict B alone, or B and S together?

Question. Flow matching (FM) from the mixture M : predict only the background and read $\hat{S} = M - \hat{B}$, or predict both components? And what does training each mixture with its *own* background (simulation truth) change? Judged on individual jets, not only on the average energy resolution.

Controlled comparison. Three FM arms with the same network (the UVCAN-S generator), the same 633k training mixtures and 2 h on one A6000, 3 seeds each; the published **UVCAN-S** (~ 105 h of training) and **Area** subtraction on the same 20k PYTHIA+HIJING and 20k JEWEL+HIJING events; the paper's jet analysis rebuilt.

Answer (PYTHIA val, $R = 0.4$, $20 < p_T < 30$ GeV):

- **No FM arm is as faithful as UVCAN-S** (scale 0.98, 3% fakes, substructure distributions $2-7\times$ closer than the best FM arm's). After a frozen calibration the jet resolution is the same for **UVCAN-S**, **unpaired FM** and **paired FM** (0.150-0.154); **joint FM** is 5-7% worse.
- **Predicting B alone loses part of the jet core**: 12-25% of the energy in towers with $S > 10$ GeV goes into $\hat{B}$. **paired FM**: jet scale 0.83. **unpaired FM** looks unbiased (0.99) only because a soft floor adds 112 GeV per event: 15% fake jets at 14-20 GeV.
- **Predicting B and S keeps the hard core** ($\leq 3\%$ lost) but drops soft signal: scale 0.91 at $R = 0.4$ (0.99 at $R = 0.2$), efficiency 0.88 at 14-20 GeV, $10\times$ larger seed spread.
- **Pairing** cuts the per-tower error $3\times$ and fakes $30\times$ and gives the smallest per-jet errors of girth, z_{lead} and r_g of all models, but does not fix the jet scale. It needs simulation truth.

Setup: one network and one budget; only the pairing and the outputs change

arm	flow	training pairs	loss	signal readout
unpaired FM	$M \rightarrow B$	minibatch exact OT: real mixtures $\leftrightarrow$ independent HIJING events	CFM MSE	$\hat{S} = M - \hat{B}$
paired FM	$M \rightarrow B$	each mixture with <i>its own</i> background ($B = M - S$, bookkeeping)	CFM MSE	$\hat{S} = M - \hat{B}$
joint FM	$(M, 0) \rightarrow (B, S)$	each mixture with its own (B, S)	$\text{mean}_B + \text{mean}_S$, weight 1 each	direct $\hat{S}$; $M - \hat{B}$ (reported separately)

Common to the three FM arms

- velocity net: the UVCGAN-S generator (ViT-ModNet, 32.3M parameters), time added to its style token
- $\log(E + 0.1)$ standardised; straight path, $\sigma = 0$; deterministic transports, not samplers
- Adam $2 \cdot 10^{-4}$, batch 256, EMA 0.9999, **2 h on one A6000**, 3 seeds (32k updates)
- the same 633k real embedded training mixtures
- one checkpoint per run: lowest val cone resolution (4 Euler steps, $M - \hat{B}$)
- solves: 4 Euler steps and midpoint 32 NFE (= 64 NFE on val); each readout shown at its val-preferred solve, both in the backup

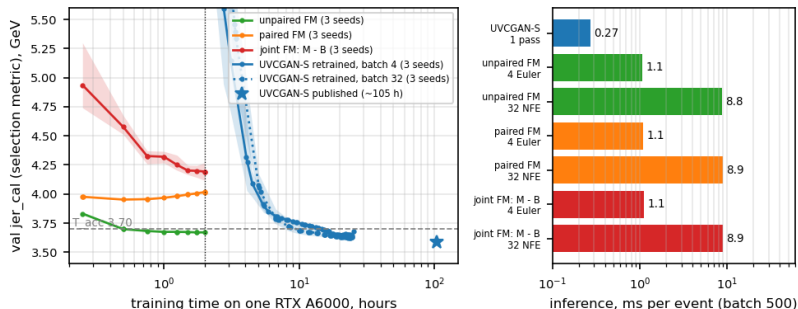
References on the same events

- **UVCGAN-S** published, one pass: batch 4, 800k updates (~ 105 h); GAN + cycle + idt-aa, i.e. also *supervised* on synthetic sums $B + S$
- **Area**: FastJet $p_T - \rho A$, no substructure. ICS unavailable (no fjcontrib)

Evaluation

- development: 20k PYTHIA+HIJING val; frozen test: 20k JEWEL+HIJING
- paper's analysis rebuilt: FastJet anti- k_T , $R = 0.2/0.4/0.5$, towers as constituents, one-to-one matching $\Delta R < 0.75R$, soft drop $z_{cut} 0.1$, $\beta 0$
- raw outputs primary; frozen calibration from val events 10k-20k
- **paired FM** and **joint FM** use simulation truth: a supervised reference, not an upper bound

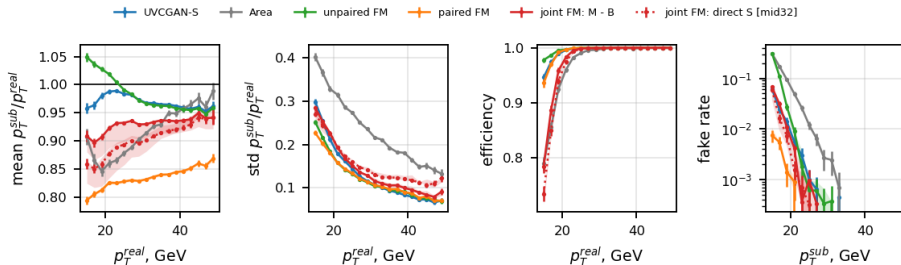
Cost: 2 h per flow on one GPU; only unpaired FM reaches 3.70 GeV



	best val res.	at 2 h	h to 3.70	updates/s	peak GB	ms/event, 4 NFE	32 NFE
unpaired FM	3.67±0.01	3.67±0.01	0.5/0.5/0.5	4.45±0.03	19.1	1.08	8.8
paired FM	3.95±0.00	4.02±0.01	-/-/-	4.55±0.03	18.1	1.10	8.9
joint FM	4.15±0.04	4.19±0.08	-/-/-	4.52±0.02	18.1	1.11	8.9
UVCGAN-S retrained, batch 4	3.64±0.02	7.44±0.70	15/17/17	—	—	—	—
UVCGAN-S retrained, batch 32	3.62±0.03	8.29±0.24	8.4/8.6/15	—	—	—	—
UVCGAN-S published (~105 h)	3.59	—	—	—	—	0.27 (1 pass)	—

One RTX A6000 each; flows 2 h, **UVCGAN-S** retrained 24 h. val res.: the selection metric (calibrated cone resolution, GeV), seed mean ± half range. h to 3.70 (T_acc = published + 3%): first confirmed checkpoint, per seed. Inference: batch 500. The joint arm's second output costs <1%.

Jets: the background-only flows lose energy or hide the loss under fakes



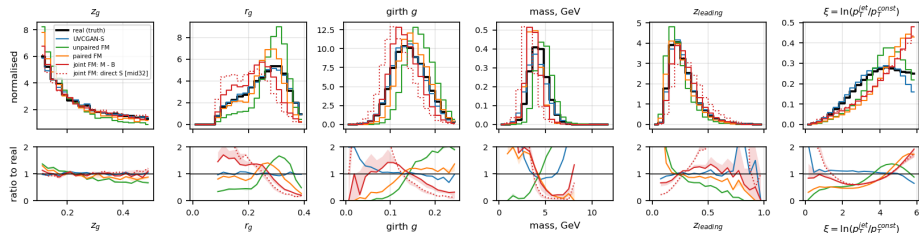
	scale	resolution	RMSE, GeV	res. (frozen cal.)	eff. 14-20	fake 14-20
UVCAN-S	0.982	0.144	3.59	0.153	0.972	0.030
Area	0.881	0.269	7.37	0.261	0.865	0.192
unpaired FM	0.993(1)	0.133	3.33(1)	0.150	0.988	0.146(1)
paired FM	0.828(2)	0.133	5.51(5)	0.154	0.968(1)	0.005
joint FM, $M - \hat{B}$	0.912(16)	0.153(1)	4.41(19)	0.161(3)	0.881(6)	0.029(3)
joint FM, direct $\hat{S}$ (32)	0.884(22)	0.163(1)	5.02(35)	0.164(1)	0.846(6)	0.020(4)
stat. error	0.001	0.001	0.03	0.001	0.004	0.003

Matched jets, $20 < p_T^{real} < 30$ GeV; efficiency (p_T^{real}), fake rate (p_T^{sub}), GeV. Raw, 4 Euler steps; (32): 32 NFE. Seed mean of 3; (n): seed half range in the last digit.

The radius separates the failure modes. Scale at $R = 0.2 / 0.4 / 0.5$: **UVCAN-S** 0.99 / 0.98 / 0.98; **unpaired FM** 0.88 / 0.99 / 1.10 (its floor fills larger cones; 67% fakes at $R = 0.5$); **paired FM** 0.82 / 0.83 / 0.85; **joint FM** 0.99 / 0.91 / 0.89 (core kept, periphery lost).

Substructure: UVCAN-S matches the distributions, paired FM the single jets

Fig. 6: substructure, $R = 0.4$, $20 < p_T^{real} < 30$ GeV, PYTHIA+HIJING (val) (ratio band: seed range)

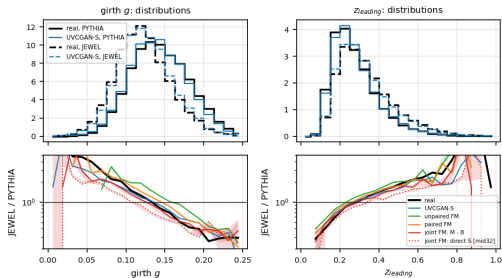


	per-jet RMSE / σ_{truth}						distribution W_1 / σ_{truth}					
	g	m	z_{lead}	p_T^D	z_g	r_g	g	m	z_{lead}	p_T^D	z_g	r_g
UVCAN-S	0.60	1.16	0.38	0.46	1.12	0.94	0.07	0.21	0.04	0.06	0.01	0.02
unpaired FM	0.79	1.31	0.48	0.72	1.12	0.97	0.59	0.68	0.35	0.60	0.22	0.38
paired FM	0.55	1.17	0.34	0.51	1.12	0.85	0.22	0.53	0.14	0.36	0.13	0.13
joint FM, $M - \hat{B}$	0.68	1.31	0.44	0.47	1.18	1.04	0.39	0.76	0.22	0.17	0.02	0.50
joint FM, direct $\hat{S}$ (32)	0.99	1.67	0.81	0.87	1.22	1.23	0.79	1.30	0.64	0.68	0.05	0.77
truth vs truth							0.03	0.03	0.02	0.02	0.01	0.03

$R = 0.4$, $20 < p_T^{real} < 30$ GeV, lower is better; σ_{truth} : spread of all truth jets. Seed half range ≤ 0.01 (background-only), ≤ 0.14 (joint); stat. error ≤ 0.010 .

JEWEL (frozen test): every method keeps 60-80% of the girth and z_{lead} shift

Fig. 7: PYTHIA against JEWEL, $R = 0.4$, $20 < p_T^{real} < 30$ GeV



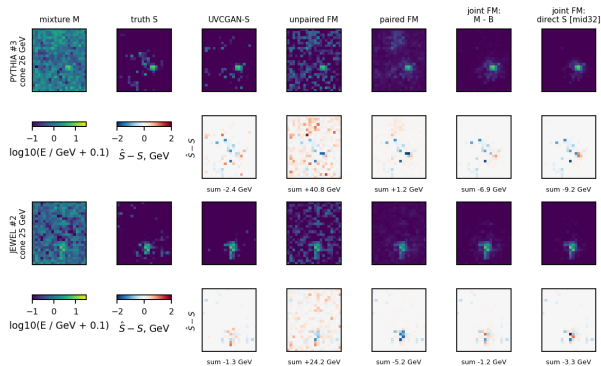
- The PYTHIA calibration carries over with a +4% shift for **UVCAN-S** and the background-only flows, +7-8% for **joint FM**; calibrated resolution stays at 0.151-0.161.
- **joint FM** ($M - \hat{B}$) does better on JEWEL than on PYTHIA (scale 0.98, per-jet girth error the smallest), consistent with its keeping hard towers: JEWEL jets are narrower.
- **unpaired FM**: fakes rise to 19%. **paired FM** keeps 0.72 of the girth shift (**UVCAN-S** 0.76), the others 0.63; the mass shift is kept least (0.22-0.55).

	scale	scale _{cal}	res. _{cal}	fake 14-20	g RMSE/ σ	z_{lead} RMSE/ σ	Δg kept	Δz_{lead} kept
UVCAN-S	1.022	1.046	0.151	0.080	0.63	0.34	0.76	0.80
Area	0.904	1.016	0.269	0.231	—	—	—	—
unpaired FM	1.033(1)	1.040	0.151(1)	0.189(2)	1.02	0.53	0.63	0.71
paired FM	0.863(2)	1.040	0.158(1)	0.015(1)	0.66	0.36	0.72	0.80
joint FM , $M - \hat{B}$	0.981(14)	1.072(1)	0.154(1)	0.075(12)	0.57(2)	0.35(1)	0.63(1)	0.73(1)
joint FM , direct $\hat{S}$ (32)	0.963(18)	1.079(1)	0.161(2)	0.062(11)	0.84(5)	0.61(4)	0.63(1)	0.80(1)

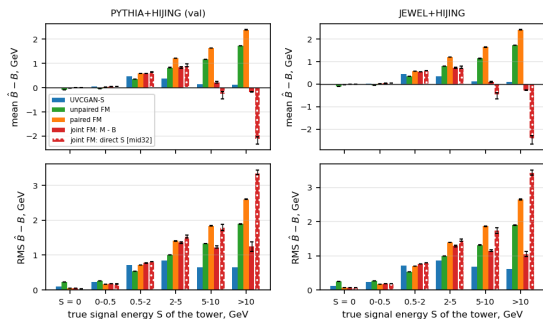
JEWEL, $R = 0.4$, $20 < p_T^{real} < 30$ GeV, raw; cal: each run's val (PYTHIA) calibration, frozen. Kept: the JEWEL–PYTHIA shift of the mean, as a fraction of the truth's (stat. error 0.03, 0.04). seed mean of 3; (n): seed half range in units of the last digit, shown where ≥ 1 .

Where the error goes: background-only flows put part of the jet core into $\hat{B}$

Event displays (raw outputs): 20 towers in ϕ around the jet, full η (24 towers)



Background error by true signal energy of the tower ($\hat{B} = M - \hat{S}$; seed mean, whiskers: seed range)



	MAE	% S lost, 5-10	>10 GeV	$\sum(\hat{S} - S)$
UVCAN-S	0.033	2	1	-7
unpaired FM	0.150	17	12	112
paired FM	0.050	24	16	8
joint FM, $M - \hat{B}$	0.031	3	-1	-19
joint FM, direct $\hat{S}$ (32)	0.031	-4	-14	-25

val, raw; per tower (GeV), share of the true signal of towers with S

in 5-10 / >10 GeV put into $\hat{B}$, GeV per event; seed mean.

In towers with $S < 2$ GeV every model misses 20-60% of the signal, **UVCAN-S**

included (the 4-step **unpaired FM** floor over-fills the softest).

Conclusions

- **Average resolution hides the failure modes.** **unpaired FM** has the best cone resolution of the flows (3.67 GeV) and a jet scale of 0.99, yet puts 12% of the hardest towers' energy into the background and adds a soft floor: 15% fakes and the worst substructure distributions.
- **Background-only vs joint (both paired, same budget): neither dominates.** Background-only keeps soft signal and gives the best per-jet substructure but loses 16-23% of the hard core (scale 0.83 / 0.75 by solver). Joint keeps the hard core but loses soft signal and low- p_T jets, and varies more between seeds. With source $(M, 0)$ its signal channel reveals the answer for $t > 0$: the flow is decided in its first step (signal-channel loss 0.51 at $t = 0$, 0.0015 at $t = 0.02$).
- **Pairing is not the fix for the core loss.** It improves images, fakes and per-jet substructure; the jet scale of background-only prediction stays low. It also needs per-mixture simulation truth.
- **UVCAN-S remains the reference for fidelity**, after ~ 105 h and with its own supervised term. The flows train in 2 h and match its calibrated jet resolution, not its jet scale or distributions.
- **Limits:** one network and 2-h budget for the flows; the analysis is rebuilt from the paper (no public code); ICS missing; **UVCAN-S** saw the val signal images in training (unpaired); the JEWEL sample's generator settings are not recorded.

Backup: both solves for every readout (same checkpoints)

	solve	cone res.	JEWEL	scale	res. _{cal}	fake	g RMSE/ σ	% $S > 10$	MAE
unpaired FM	4 Euler	3.67 ± 0.01	3.55 ± 0.01	0.993(1)	0.150	0.146(1)	0.79	12	0.150
	32 NFE	4.10 ± 0.01	3.65 ± 0.01	0.742(2)	0.155(1)	0.002	0.53	24	0.051
paired FM	4 Euler	3.95 ± 0.00	3.61 ± 0.00	0.828(2)	0.154	0.005	0.55	16	0.050
	32 NFE	4.11 ± 0.01	3.66 ± 0.01	0.751(2)	0.157	0.002	0.54	23	0.051
joint FM , $M - \hat{B}$	4 Euler	4.15 ± 0.04	4.12 ± 0.05	0.912(16)	0.161(3)	0.029(3)	0.68(4)	-1	0.031
	32 NFE	4.30 ± 0.05	4.32 ± 0.08	0.877(23)	0.165(3)	0.024(6)	0.95(6)	-3	0.030
joint FM , direct $\hat{S}$	4 Euler	5.80 ± 0.25	5.19 ± 0.19	0.780(18)	0.174(5)	0.004(1)	0.86(4)	-13	0.033
	32 NFE	4.59 ± 0.08	4.40 ± 0.09	0.884(22)	0.164(1)	0.020(4)	0.99(6)	-14	0.031

Cone res.: calibrated resolution in the true $R = 0.4$ cone, GeV (the selection metric), val and JEWEL, seed mean $\pm$ half range. Jets: $R = 0.4$, 20-30 GeV; fake 14-20 GeV; % $S > 10$: energy of towers with $S > 10$ GeV put into $\hat{B}$; MAE per tower, GeV. Seed 0 at 64 NFE, val cone res.: **unpaired FM** 4.10, **paired FM** 4.10, **joint FM**, $M - \hat{B}$ 4.34, **joint FM**, direct $\hat{S}$ 4.52. Joint $\hat{B} + \hat{S} - M$ before clean-up (val): 4 Euler: 0.009 GeV per tower (MAE), -7.7 GeV per event; 32 NFE: 0.003 GeV per tower (MAE), +1.3 GeV per event.

- Val prefers 4 Euler steps for every readout except the joint direct $\hat{S}$ (32 NFE); the main slides use these. The accurate solve lowers the background-only jet scale further ($0.83 \rightarrow 0.75$) and removes the unpaired floor (fakes $0.146 \rightarrow 0.002$): the floor is a 4-step artefact.
- 32 NFE is resolved for the background-only flows (64 NFE: same val cone resolution to 0.01 GeV); for the joint direct readout 64 NFE changes it by 0.1 GeV.
- Labelled clean-up (0.5 GeV tower threshold on every output) removes fakes but lowers every scale by a further 11-27%; it does not restore the core (bench_summary.csv, [thr0.5]).

Backup: comparability, limitations, what is missing

Comparability

- Training time: flows 2 h (32k updates at batch 256); **UVCAN-S** ~ 105 h (800k at batch 4). The equal-time view is **UVCAN-S** retrained from scratch: 8.4-16.7 h to 3.70 GeV, 6.9-8.6 GeV at 2 h.
- Information: **paired FM**, **joint FM** use each training mixture's true background; **UVCAN-S**'s `idt-aa` uses synthetic sums of independent B and S ; **unpaired FM** uses no pairing. Same 633k mixtures for all flows.
- One checkpoint per run, selected on val only (cone resolution, 4 Euler, $M - \hat{B}$); JEWEL never used for a choice. The paired arm's selected checkpoints are early (30-45 min): its cone resolution worsens with training while its per-tower error improves.
- Statistical errors (one run) are shown apart from the seed spread (3 seeds).
- Supplementary true-axis cone scores (`bench_cone*.csv`): same ordering; the shape part of the EMD is smallest for **paired FM** (2.93 GeV val) vs **UVCAN-S** 3.12.

Missing or limited

- ICS baseline: not available (fjcontrib not installed).
- No public analysis code: tower threshold (> 0), greedy matching, fake rate binned in p_T^{sub} , C/A soft-drop reclustering, jets > 5 GeV are our choices; Figs. 3-7 reproduced (`docs/flow/bench`), no paper numbers to compare.
- The val events' signal images are among **UVCAN-S**'s (unpaired) training signals; HIJING parents of the embedded mixtures cannot be checked for train/val overlap (mixture keys are disjoint).
- JEWEL sample (Zenodo 17594612): images only, generator version and settings not recorded (the paper states JEWEL 2.3.0 + Geant4).
- Joint FM is still improving at 2 h; the budget is the same for all flows by design.

Follow-up: does a noisy training path stop the paired flow's core leakage?

The failure. **paired FM** background-only FM puts part of the jet core into $\hat{B}$ although every training mixture comes with its own background: jet scale 0.83 (4 Euler) / 0.75 (accurate solve); 16% / 23% of the signal energy of towers with $S > 10$ GeV ends in $\hat{B}$. With $\hat{S} = M - \hat{B}$, lost signal means $\hat{B}$ is *too high*: the flow stops short of the true background in the core.

Hypothesis (not an established explanation). Noiseless straight interpolants supervise the velocity only on the segments between $(z(M), z(B))$ pairs; the states the ODE visits drift off them. A noisy interpolant that returns exactly to the endpoints trains the field around the paths. Caveat: an exactly learned marginal transport need not be the best estimator for single events.

Path (stochastic interpolant, Albergo, Boffi & Vanden-Eijnden 2025), $a = z(M)$, $b = z(B)$:

$$x_t = (1-t)a + tb + \eta \sin(\pi t) \varepsilon, \quad u_t = (b-a) + \eta \pi \cos(\pi t) \varepsilon,$$

the same $\varepsilon \sim N(0, I)$; $\eta =$ largest noise sd ($t = \frac{1}{2}$), standardised log E units; exact endpoints. Inference unchanged: deterministic ODE from $z(M)$, $\hat{S} = M - \hat{B}$.

Controlled pilot (seed 0): $\eta = 0$ against $\eta = 0.1$; same endpoint pairs, UVCAN-S net, 633k mixtures, optimiser, EMA, readout; ε from its own RNG (same weights, mixture order, times); **32,640 updates each**, compared at the final update.

- checks: $\eta = 0 =$ old path bit for bit; exact endpoints; $u_t = \partial_t x_t$ to $3 \cdot 10^{-10}$; the fresh control repeats the benchmark run's loss curve
- primary: accurate solve (midpoint 32 NFE; 64 checked); 4 Euler shown apart

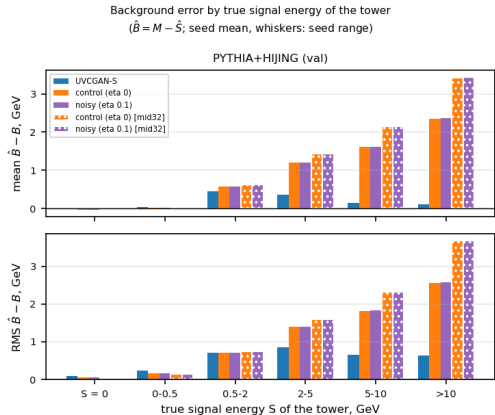
Pilot result: $\eta = 0.1$ changes nothing measurable (seed 0, 32,640 updates, val)

	32 NFE (accurate)		4 Euler	
	$\eta = 0$	$\eta = 0.1$	$\eta = 0$	$\eta = 0.1$
% of S in $\hat{B}$: $S > 10$ GeV	23.1	23.2	15.8	15.9
S 5-10 GeV	31.4	31.4	23.6	23.7
S 2-5 GeV	47.0	47.0	39.4	39.6
jet scale 20-30	0.753	0.753	0.832	0.831
resolution (frozen cal.)	0.161	0.161	0.156	0.157
efficiency 14-20	0.956	0.959	0.969	0.971
fake rate 14-20	0.002	0.002	0.006	0.005
$\hat{B} - B$ at $S = 0$, GeV	-0.025	-0.025	-0.025	-0.025
$\sum(\hat{S} - S) / \text{event}$, GeV	0.7	0.7	6.1	6.0
MAE / tower, GeV	0.050	0.050	0.049	0.049

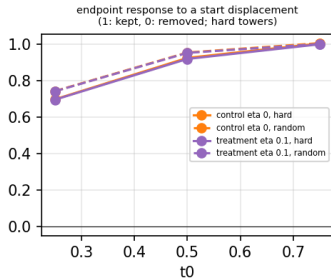
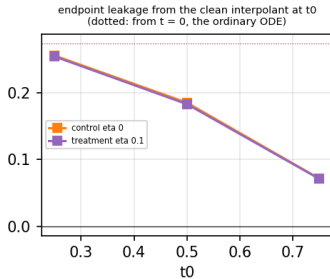
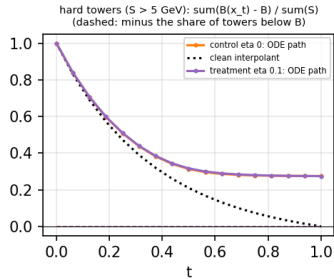
Substructure (per-jet RMSE/ σ and $W1/\sigma$ of $g, m, z_{lead}, p_T^D, z_g, r_g$): $\eta = 0$ and 0.1 agree to $\leq 0.007\sigma$ at both solves (noisy/noisy_compare.csv).

UVCGAN-S for reference: scale 0.98, 0.8% of the core in $\hat{B}$.

All 58 pre-registered comparisons are within $3\times$ the seed or statistical spread (largest core change $+0.0007$, threshold 0.0019). 64 NFE agrees with 32 to 0.0004 GeV per tower. The 4-step readout keeps its trade-off in both arms: less core in $\hat{B}$, higher scale, $+6$ GeV per event of spurious signal.



Why not: the core stays in $\hat{B}$ even when the solve starts on the true path



256 val events, 592 towers with $S > 5$ GeV, final update, midpoint. Left: the ODE never undershoots (no hard tower goes below the true B), follows the clean interpolant to $t \approx 0.2$ and stalls at 27% after $t \approx 0.7$. Middle: started exactly on the true path at t_0 , the rest of the solve still leaves 26% / 18% / 7% of the core in $\hat{B}$. Right: a displaced core is kept, not restored. Both arms agree to 0.003.

Answer. Noisy interpolation ($\eta = 0.1$) improved neither the accurately solved background nor the jets, and did not shift the numerical trade-off either: both solves are unchanged. The loss comes from the learned field on the paths, not from a lack of supervision around them (hypothesis: the regression $E[b - a | x_t]$ averages over backgrounds the partly subtracted state cannot tell apart). Pilot stopped: no seeds 1-2, no sweep, JEWEL untouched. This rules out only this setting and says nothing about unpaired vacuum-to-medium transfer.